University of Padua

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COMPREHENSIVE LECTURE NOTES

Digital Transmission and Embedded Systems

From Theoretical Signal Foundations to Hardware Conditioning and Real-Time Control

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Contents

Chapter 1 **Foundations of Signal Theory and Sampling** ____ **Page 3**

1.1 Bandlimited Signals and the Frequency Domain 3
1.2 The Nyquist-Shannon Sampling Theorem 3
1.3 Inter-Symbol Interference and Nyquist's Criterion 5

Chapter 2 **Analog Conditioning and Front-End** ____ **Page 6**

2.1 Active Anti-Aliasing Filter Design 6
2.2 Placeholder for High-Frequency Front-End Schematic 7
2.3 Practical Component Sizing Exercise 7

Chapter 3 **Digital Processing and Embedded Real-Time Implementation** ____ **Page 8**

3.1 Circular Buffer Acquisition Algorithm 8
3.2 Embedded C Implementation 8
3.3 Comparative Filter Performance Benchmark 9
3.4 Real-Time Schedulability Verification Exercise 9

1 Foundations of Signal Theory and Sampling

1.1 Bandlimited Signals and the Frequency Domain

In digital communication systems, the fundamental goal is to transmit binary information across a physical communication channel subject to finite bandwidth constraints, linear phase distortion, and additive thermal noise.

Consider a deterministic continuous-time signal $x(t) \in \mathcal{L}^2(\mathbb{R})$. Its **Fourier transform** is conventionally defined as:

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{+\infty} x(t)\, e^{-j2\pi ft}\, \mathrm{d}t \tag{1.1}$$

with the corresponding inverse synthesis transform given by:

$$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{+\infty} X(f)\, e^{j2\pi ft}\, \mathrm{d}f \tag{1.2}$$

Definition 1.1.1 (Bandlimited Signal)

A continuous-time signal $x(t)$ is said to be strictly bandlimited in the Fourier sense if there exists a finite cut-off frequency $B > 0$, termed the **nominal bandwidth**, such that its spectral energy density vanishes identically outside $[-B, B]$:

$$X(f) = 0, \quad \forall |f| > B$$

By Parseval's energy conservation theorem, the total energy E_x of a finite-energy signal is preserved identically across domains:

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2\, \mathrm{d}t = \int_{-\infty}^{+\infty} |X(f)|^2\, \mathrm{d}f = \int_{-B}^{+B} |X(f)|^2\, \mathrm{d}f \tag{1.3}$$

In practical engineering systems, strict bandlimiting ensures that $x(t)$ is smooth and analytic over $\mathbb{R}$, precluding sudden jump discontinuities in the time domain and guaranteeing strong temporal correlation between adjacent samples.

1.2 The Nyquist-Shannon Sampling Theorem

Sampling represents the fundamental bridge between the continuous physical realm and discrete digital processing. In periodic uniform sampling, a continuous signal $x(t)$ is evaluated at fixed temporal intervals T_s:

$$x[n] \overset{\text{def}}{=} x(nT_s), \quad f_s \overset{\text{def}}{=} \frac{1}{T_s} \tag{1.4}$$

where f_s is the sampling rate in Hertz. If the conversion process is performed too slowly, high-frequency spectral components alias irreversibly into the baseband, permanently destroying signal fidelity. The central question answered by Harry Nyquist (1928) and formalized by Claude Shannon (1949) is: *what is the minimal sampling frequency required to capture continuous information without irreversible degradation?*

Theorem 1.2.1 (Sampling Theorem (Nyquist-Shannon))

Let $x(t)$ be a continuous-time signal strictly bandlimited to $[-B, B]$. If the sampling frequency f_s satisfies the non-overlapping condition:

$$f_s \geq 2B \tag{1.5}$$

then the continuous signal $x(t)$ can be reconstructed **exactly and without distortion** for all $t \in \mathbb{R}$ from its discrete samples $x(nT_s)$ via cardinal sinc interpolation:

$$x(t) = \sum_{n=-\infty}^{+\infty} x(nT_s)\,\mathrm{sinc}\left(\frac{t - nT_s}{T_s}\right) \tag{1.6}$$

where $\mathrm{sinc}(u) \overset{\text{def}}{=} \dfrac{\sin(\pi u)}{\pi u}$ for $u \neq 0$ and $\mathrm{sinc}(0) = 1$.

***Proof* (Spectral Proof).** Ideal impulse sampling can be modeled as the multiplication of $x(t)$ by a periodic Dirac comb (Shah distribution) $p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT_s)$:

$$x_p(t) = x(t) \cdot \sum_{n=-\infty}^{+\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{+\infty} x(nT_s)\delta(t - nT_s)$$

Applying the Fourier property that maps a time-domain product to a spectral convolution, and recalling that $\mathcal{F}\{p(t)\} = f_s \sum_{k=-\infty}^{+\infty} \delta(f - kf_s)$:

$$X_p(f) = X(f) * P(f) = f_s \sum_{k=-\infty}^{+\infty} X(f - kf_s)$$

The sampled spectrum $X_p(f)$ consists of an infinite summation of replicas centered at integer multiples kf_s. When $f_s - B \geq B \iff f_s \geq 2B$, adjacent spectral replicas do not overlap. The baseband replica ($k = 0$) can be isolated by applying an ideal low-pass brickwall filter:

$$H_{rec}(f) = T_s \ \mathrm{rect}\left(\frac{f}{2B}\right) = \begin{cases} T_s & |f| \leq B \\ 0 & |f| > B \end{cases}$$

Multiplying $X(f) = X_p(f) \cdot H_{rec}(f)$ and transforming back to the time domain yields the classical Whittaker-Shannon cardinal series. 😊

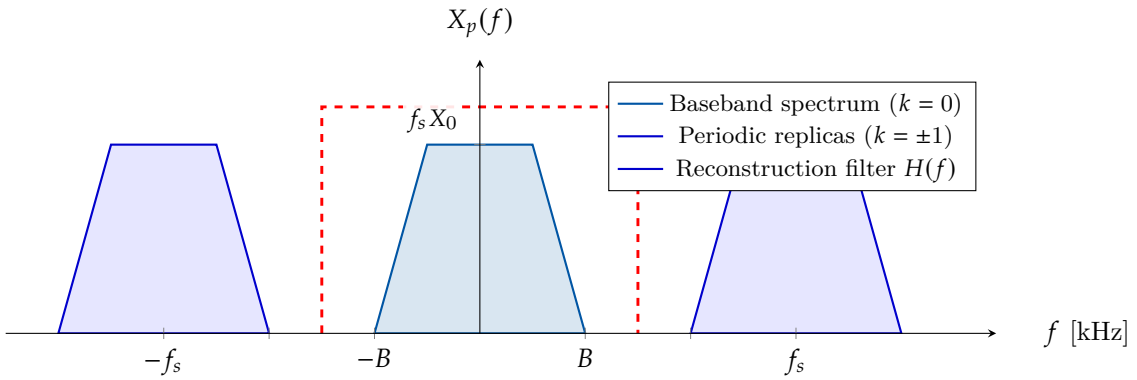


Figure 1.1: Frequency-domain representation of periodic sampling without aliasing ($f_s > 2B$).

1.3 Inter-Symbol Interference and Nyquist's Criterion

When modulated pulses propagate across a bandlimited channel, dispersive spreading causes adjacent pulse tails to overlap, generating Inter-Symbol Interference (ISI).

Formula 1.3.1 (Nyquist Criterion for Zero ISI)

To ensure that a pulse-shaping filter $g(t)$ causes no interference at optimum symbol decision instants $t = mT$, its frequency response $G(f)$ must satisfy:

$$\sum_{k=-\infty}^{+\infty} G\left(f - \frac{k}{T}\right) = T, \quad \forall |f| \leq \frac{1}{2T} \tag{1.7}$$

Remark 1.3.2 (Spectral Efficiency vs. Jitter Sensitivity)

The theoretical minimum-bandwidth pulse satisfying (1.7) is the sinc pulse, achieving $B_{min} = \frac{1}{2T} = \frac{R_s}{2}$ Hz. However, because sinc pulses decay as slowly as $1/t$, small timing offsets or clock jitter induce divergent cumulative ISI series.

Example 1.3.3 (Raised Cosine Spectrum). To ensure rapid temporal decay without infinite tails, communication systems employ the **raised cosine** filter with roll-off factor $\alpha \in [0, 1]$. The total transmission bandwidth is given by:

$$B = \frac{1+\alpha}{2T} = \frac{R_s}{2}(1+\alpha)$$

For a link operating at $R_s = 100$ kbaud with roll-off $\alpha = 0.25$, the required channel bandwidth is $B = \frac{100}{2}(1.25) = 62.5$ kHz.

2 Analog Conditioning and Electronic Front-End

2.1 Active Anti-Aliasing Filter Design

Before an analog signal can be safely converted by an ADC, an analog anti-aliasing filter must attenuate spectral noise and high-frequency interferers above the Nyquist frequency $f_s/2$.

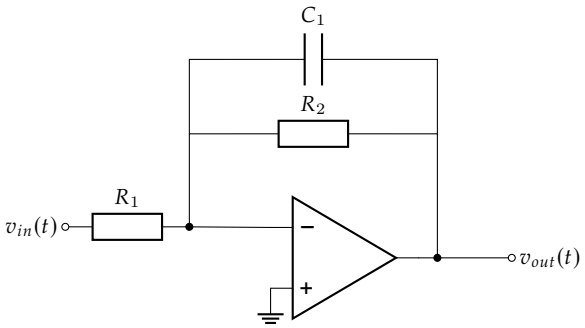


Figure 2.1: First-order inverting active low-pass filter stage for ADC interface.

Nodal analysis at the inverting virtual ground yields the Laplace transfer function:

$$W(s) = \frac{V_{out}(s)}{V_{in}(s)} = -\frac{Z_2(s)}{Z_1(s)} = -\frac{R_2 \parallel \frac{1}{sC_1}}{R_1} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + sR_2C_1} \tag{2.1}$$

Proposition 2.1.1 (Cut-Off Frequency and Passband Gain)

Defining $A_0 \overset{\text{def}}{=} -\frac{R_2}{R_1}$ (DC passband gain) and $\omega_H \overset{\text{def}}{=} \frac{1}{R_2C_1}$, the frequency response magnitude is:

$$|W(j\omega)| = \frac{|A_0|}{\sqrt{1 + (\omega/\omega_H)^2}} \tag{2.2}$$

At the cut-off angular frequency $\omega = \omega_H$, the gain drops by $\sqrt{2}$, corresponding to the conventional -3 dB attenuation threshold.

Warning 2.1.2 (Saturation and Supply Rails)

Real operational amplifiers cannot swing beyond their power supply voltages:

$$V_{EE} + V_{sat} \le v_{out}(t) \le V_{CC} - V_{sat}$$

Driving the amplifier beyond these limits induces severe harmonic clipping, producing spurious high-frequency harmonics that fold back into the baseband spectrum as in-band aliasing.

Note 2.1.3 (Single-Supply DC Biasing)

In modern unipolar embedded systems ($0-3.3$ V microcontrollers), the non-inverting terminal must be biased at $V_{ref} = V_{DD}/2 = 1.65$ V via a decoupled resistive voltage divider to center the dynamic range.

2.2 Placeholder for High-Frequency Front-End Schematic

Lecture slides and hardware datasheets often contain complex multi-stage diagrams that must be cropped and integrated during revision:

[Figure to insert]

Slide 22 (Prof. Rossi): Superheterodyne receiver front-end block diagram with I/Q demodulation mixer and Programmable Gain Amplifier (PGA)

2.3 Practical Component Sizing Exercise

Exercise 2.3.1 (Anti-Aliasing Filter Component Sizing). Design the active first-order low-pass filter shown in Figure 2.1 to interface a seismic sensor to an ADC running at $f_s = 20$ kHz. The design specifications are:

a. Half-power bandwidth (-3 dB cut-off) at $f_H = 4$ kHz.

b. In-band passband voltage gain magnitude $|A_0| = 5$ (14 dB).

c. Input resistance $R_1 \geq 10$ kΩ to prevent loading effects on the transducer.

Select standard commercial values (E12 series) for R_1, R_2, and C_1.

Solution.

a. **Selection of R_1 and R_2**: Setting the input resistance to $R_1 = 10$ kΩ to satisfy the source loading constraint, the feedback resistance required for a gain $|A_0| = R_2/R_1 = 5$ is:

$$R_2 = 5 \cdot 10 \text{ k}\Omega = 50 \text{ k}\Omega \xrightarrow{\text{E12 series}} R_2 = 47 \text{ k}\Omega$$

giving an effective passband gain $|A_{0,eff}| = 4.7$ (less than 6% design deviation).

b. **Capacitance Calculation for C_1**: From the cut-off frequency relationship $\omega_H = 2\pi f_H = 1/(R_2 C_1)$:

$$C_1 = \frac{1}{2\pi f_H R_2} = \frac{1}{2\pi \cdot 4000 \cdot 47 \times 10^3} \approx 846 \text{ pF} \xrightarrow{\text{E12 series}} C_1 = 820 \text{ pF}$$

■

3 Digital Processing and Embedded Real-Time Implementation

3.1 Circular Buffer Acquisition Algorithm

In real-time microcontroller architectures (e.g., ARM Cortex-M or RISC-V), ADC conversions are triggered by periodic hardware timers and transferred into memory via DMA or fast interrupts. To decouple acquisition rates from computation deadlines without introducing data races, a lock-free circular ring buffer is utilized.

Algorithm 1: Real-time causal FIR filtering on a circular ring buffer.

Input: New ADC conversion sample x_new, write index pointer w_idx
Output: Updated buffer B[0...N-1], computed filter output y

```
B[w_idx] ←x_new;
w_idx ←(w_idx + 1) mod N acc ← 0 for k ←0 to M - 1 do
    idx ←(w_idx - 1 - k + N) mod N acc ← acc + h[k] * B[idx]
return acc;
```

3.2 Embedded C Implementation

The listing below illustrates the production C Interrupt Service Routine (ISR) executing on a 32-bit embedded target:

```c
#include <stdint.h>
#define BUFFER_SIZE   64
#define TAPS_COUNT    8

static volatile uint16_t adc_buffer[BUFFER_SIZE];
static volatile uint8_t  head_idx = 0;
// Symmetric FIR coefficients (fixed-point Q15 representation)
static const int16_t fir_coeffs[TAPS_COUNT] = {
    1024, 3450, 7812, 11420, 11420, 7812, 3450, 1024
};

void ADC_Timer_IRQHandler(void) {
    // 1. Hardware sample acquisition & circular buffer insert
    adc_buffer[head_idx] = ADC1->DR;
    head_idx = (head_idx + 1) & (BUFFER_SIZE - 1); // Fast power-of-two wrap

    // 2. Compute discrete convolution sum
    int32_t accumulator = 0;
    for (int k = 0; k < TAPS_COUNT; ++k) {
        uint8_t sample_idx = (head_idx - 1 - k) & (BUFFER_SIZE - 1);
        accumulator += (int32_t)fir_coeffs[k] * adc_buffer[sample_idx];
    }

    // 3. Output scaling to unipolar DAC range
    int16_t filtered = (int16_t)(accumulator >> 15);
    DAC1->DHR12R1 = (uint16_t)(filtered + 2048);
}
```

Listing 3.1: Embedded C timer ISR for periodic ADC sampling and FIR filtering.

3.3 Comparative Filter Performance Benchmark

Table 3.1 summarizes the computational trade-offs measured on an ARM Cortex-M4 microcontroller running at 168 MHz across digital filter topologies.

Filter Architecture	Order	CPU Cycles/Sample	Latency [μs]	Stability
Simple Moving Average (SMA)	8	18	0.11	Unconditional
Raised Cosine FIR	16	42	0.25	Unconditional
Raised Cosine FIR	32	85	0.51	Unconditional
Butterworth IIR (Biquad)	4 (2 biquads)	24	0.14	Conditional
Chebyshev Type I IIR	4 (2 biquads)	26	0.15	Conditional

Table 3.1: Complexity and execution latency across digital filter topologies.

3.4 Real-Time Schedulability Verification Exercise

Exercise 3.4.1 (Real-Time Deadline Schedulability Analysis). Consider the embedded ISR routine defined in Listing 3.1 executing on a processor clocked at $f_{cpu} = 48$ MHz. Given the following system constraints:

- The sampling frequency is set to $f_s = 50$ kHz.
- Computational filtering requires 140 core CPU execution cycles.
- Hardware interrupt context-switching overhead (prologue and epilogue) is 24 cycles.

Compute the CPU utilization factor (U) and determine whether the hard real-time deadline is strictly met.

Solution. The total period available between successive timer interrupts is:

$$T_s = \frac{1}{f_s} = \frac{1}{50\,000 \text{ Hz}} = 20\,\mu\text{s}$$

The total cycle count consumed by the processor per sample is:

$$C_{tot} = C_{calc} + C_{overhead} = 140 + 24 = 164 \text{ cycles}$$

The physical execution time is:

$$T_{exe} = \frac{C_{tot}}{f_{cpu}} = \frac{164}{48 \times 10^6 \text{ s}^{-1}} \approx 3.417\,\mu\text{s}$$

The CPU utilization factor is given by:

$$U = \frac{T_{exe}}{T_s} = \frac{3.417\,\mu\text{s}}{20\,\mu\text{s}} \approx 0.1708 \implies 17.08\%$$

Because $U < 100\%$ (and well below the classical Liu-Layland real-time bound of 69%), the system is **fully schedulable**, leaving more than 82% of CPU bandwidth available for background application processing. ■